



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

AGRICULTURAL TECHNOLOGY
MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Agricultural Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept other relevant, correct answers.

QUESTION 1

- 1.1 1.1.1 B ✓✓ 1.1.2 B ✓✓ 1.1.3 A ✓✓ 1.1.4 C ✓✓ 1.1.5 C ✓✓ (10)
- 1.2 1.2.1 Alloy ✓✓ 1.2.2 Straining post / end post ✓✓ 1.2.3 Borehole ✓✓ 1.2.4 Welding ✓✓ 1.2.5 Electric fence ✓✓ (10)
- [20]**

QUESTION 2

- 2.1 Any FIVE: wear protective clothing (overalls, safety boots, goggles) ✓; keep the workshop clean and tidy ✓; use tools only for their intended purpose ✓; switch off machines before adjusting them ✓; know where the fire extinguisher and first-aid kit are ✓; no loose clothing or jewellery near machines; never work alone with dangerous machines; ventilate the workshop (5)
- 2.2 Ferrous metals contain iron ✓, e.g. steel or cast iron ✓. Non-ferrous metals contain no iron ✓, e.g. copper, aluminium or zinc ✓. (4)
- 2.3 Iron in steel reacts with oxygen and water (moisture) ✓✓ to form rust (iron oxide). Prevention, any TWO: paint ✓; galvanising (zinc coating) ✓; oil or grease; keep dry; use stainless steel (4)
- 2.4 Cutting, any TWO: hacksaw ✓; tin snips ✓; bolt cutter; angle grinder (power tool). Measuring, any ONE: steel rule / tape measure / vernier caliper / square ✓ (3)
- 2.5 Any FOUR: clean tools after use ✓; oil metal parts to prevent rust ✓; store tools in a dry, organised place ✓; sharpen cutting tools ✓; replace or repair damaged handles; check tools before use (4)
- [20]**

QUESTION 3

- 3.1 Any THREE: keep livestock in and predators or stock thieves out ✓; divide the farm into camps for rotational grazing ✓; protect crops from animals ✓; mark boundaries; keep animals off roads (3)
- 3.2.1 $2 \times (400 + 250) \checkmark = 1\,300\text{ m} \checkmark$ (2)
- 3.2.2 $1\,300 \times 5 \checkmark = 6\,500\text{ m} \checkmark$ (2)
- 3.2.3 $1\,300 \div 5 \checkmark = 260\text{ droppers} \checkmark$ (2)
- 3.2.4 $6\,500 \div 500 = 13\text{ rolls} \checkmark$; $13 \times \text{R}1\,250 \checkmark = \text{R}16\,250 \checkmark$ (3)
- 3.3 Any logical steps: mark the position ✓; dig a deep hole ✓; place the post upright (check with a spirit level) and set it in concrete or ram the soil firmly ✓; fit a stay (brace) to resist the pull of the wires ✓. (4)

- 3.4 Any FOUR: bricks / blocks ✓; cement and concrete ✓; steel / corrugated iron ✓; wood (poles, timber) ✓; stone; thatch (4)

[20]

QUESTION 4

- 4.1 Any FOUR, 1 mark for the source and 1 for the type: solar ✓ renewable ✓; wind ✓ renewable ✓; diesel / petrol ✓ non-renewable ✓; Eskom electricity from coal ✓ non-renewable ✓; biogas (renewable); hydropower (renewable); wood (renewable) (8)
- 4.2.1 $750 \text{ W} = 0,75 \text{ kW} \checkmark$; $0,75 \times 6 \checkmark = 4,5 \text{ kWh} \checkmark$ (3)
- 4.2.2 $4,5 \times 30 = 135 \text{ kWh} \checkmark$; $135 \times \text{R}2,50 \checkmark = \text{R}337,50 \checkmark$ (3)
- 4.3 Any THREE: free energy after installation ✓; works during load shedding / in remote areas without the grid ✓; renewable and does not pollute ✓; low maintenance (3)
- 4.4 Any ONE rule, explained, e.g.: switch off the power at the main switch before working on a circuit ✓✓, because electricity can cause serious burns or death ✓. (Others: use qualified electricians; keep electrical equipment dry; do not overload plugs; check cables for damage.) (3)

[20]

QUESTION 5

- 5.1 Any FOUR: boreholes ✓; rivers and streams ✓; dams ✓; rainwater harvested in tanks ✓; springs; municipal supply (4)
- 5.2 Any TWO: windmill (wind pump) ✓; electric submersible pump ✓; solar pump; hand pump; centrifugal pump; diesel pump (2)
- 5.3 $r = 2 \div 2 = 1 \text{ m} \checkmark$; $V = \pi r^2 h \checkmark = 3,14 \times 1^2 \times 1,5 \checkmark = 4,71 \text{ m}^3 \checkmark = 4\,710 \ell \checkmark$ (5)
- 5.4 Any FOUR: drip irrigation ✓; irrigate early in the morning or evening ✓; fix leaking pipes and troughs ✓; collect rainwater ✓; use mulch; float valves in troughs; recycle water (4)
- 5.5 Water can contain harmful bacteria or parasites ✓ that make animals sick ✓; it can contain too much salt, nitrates or other chemicals ✓ that reduce growth and production ✓; regular testing finds problems early so they can be treated ✓. (5)

[20]

TOTAL: 100

END OF MARKING GUIDELINE